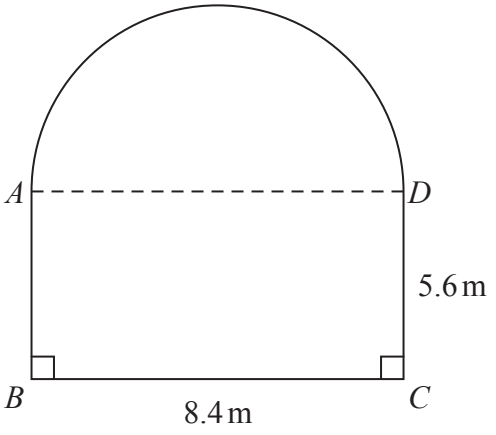
24 A garden is in the shape of a rectangle, *ABCD*, and a semicircle.
AD is the diameter of the semicircle.



Carol is going to cover the garden with fertiliser.

A box of fertiliser costs £4.99
Carol has been told that one box of fertiliser will cover 12 m^2 of garden.

(a) Work out the cost of buying enough fertiliser to cover the garden completely.

Radius = 8.4 ÷ 2 = 4.2 m; full circle area = 17.64π m^2.

Angle	Area (m^2)
360	17.64π
180	8.82π

1 mark
π × 4.2^2, then half

Semicircle area = 8.82π m^2.

1 mark
Add rectangle and semicircle

Garden = 8.4 × 5.6 + 8.82π ≈ 74.7488 m^2.

1 mark
Round boxes up to 7

74.7488... ÷ 12 = 6.229...; buy 7 boxes.

1 mark
Multiply by box cost

Cost = 7 × £4.99 = £34.93.

1 mark
£34.93

£ .....

(5)

Carol finds out that one box of fertiliser will cover more than 12 m^2 of garden.

(b) Explain how this might affect the number of boxes she needs to buy.

She might need fewer boxes, since each box covers more area.

1 mark
May need fewer boxes

(1)

(Total for Question 24 is 6 marks)